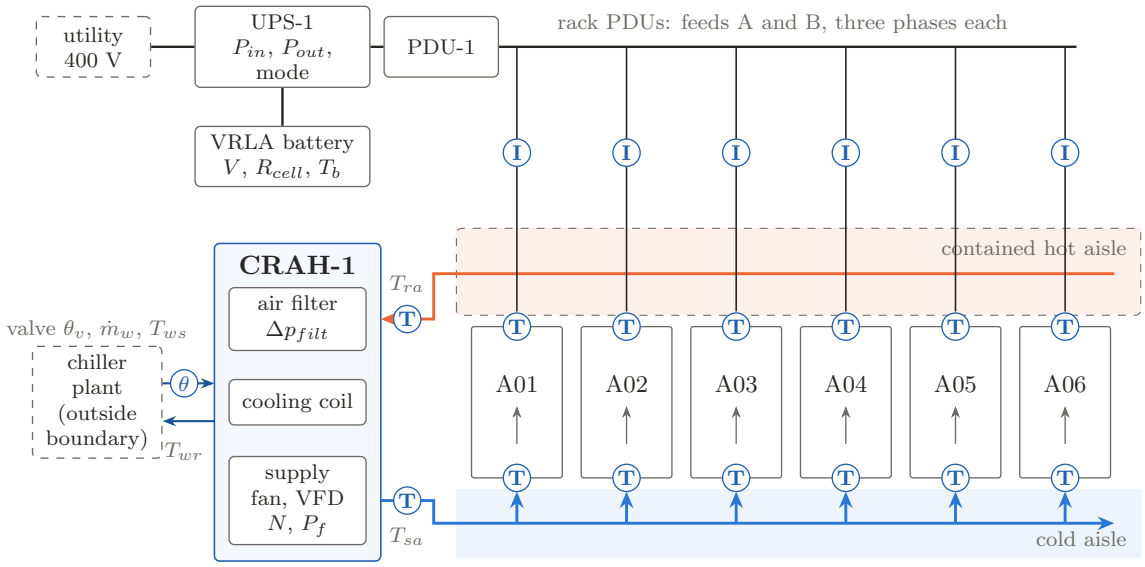
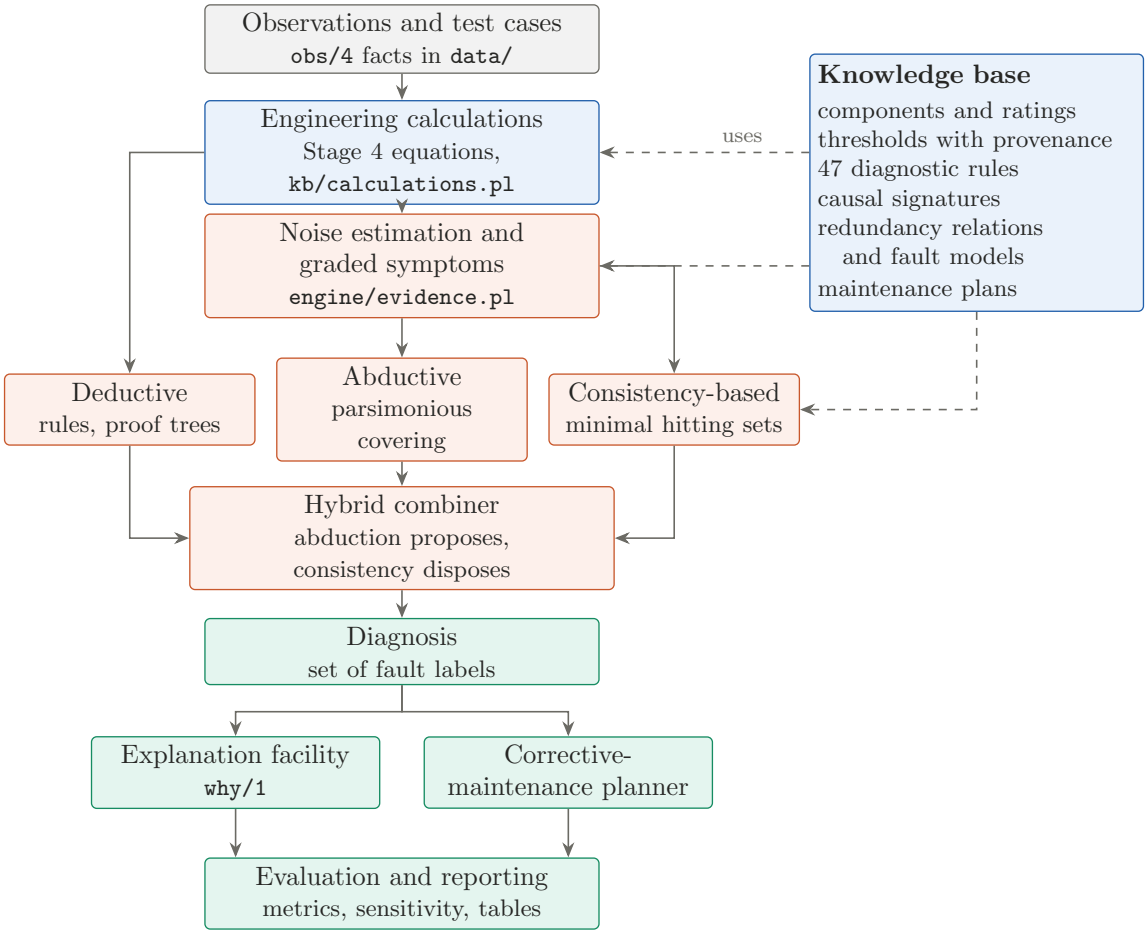
fig_system_schematic

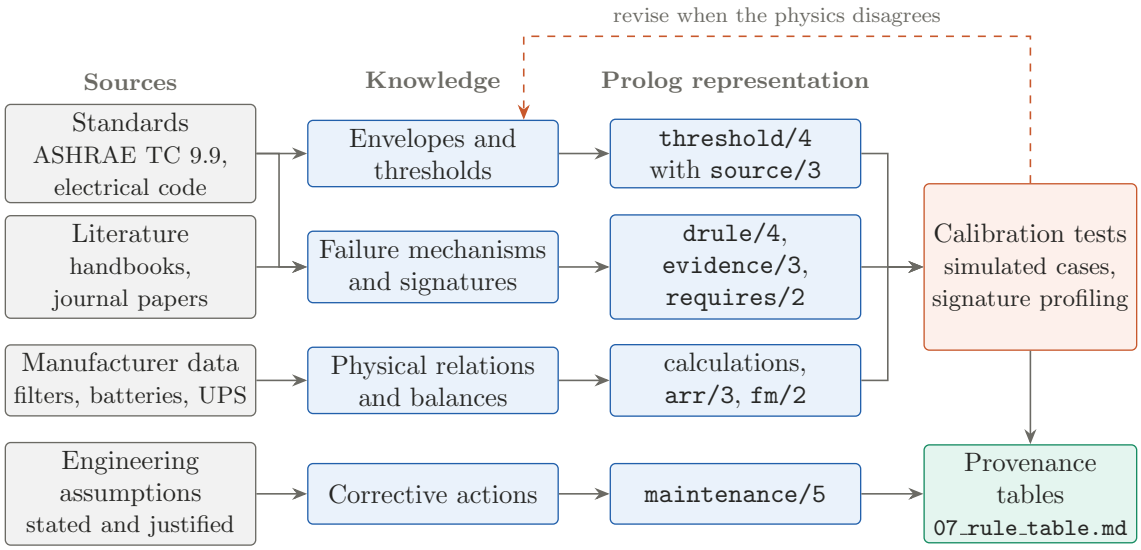


T temperature (inlet middle and top at the front of each rack, outlet mean and top at the rear); I phase currents; blue arrows supply air, orange arrows return air. Each server also reports CPU temperature, clock and fan speed.

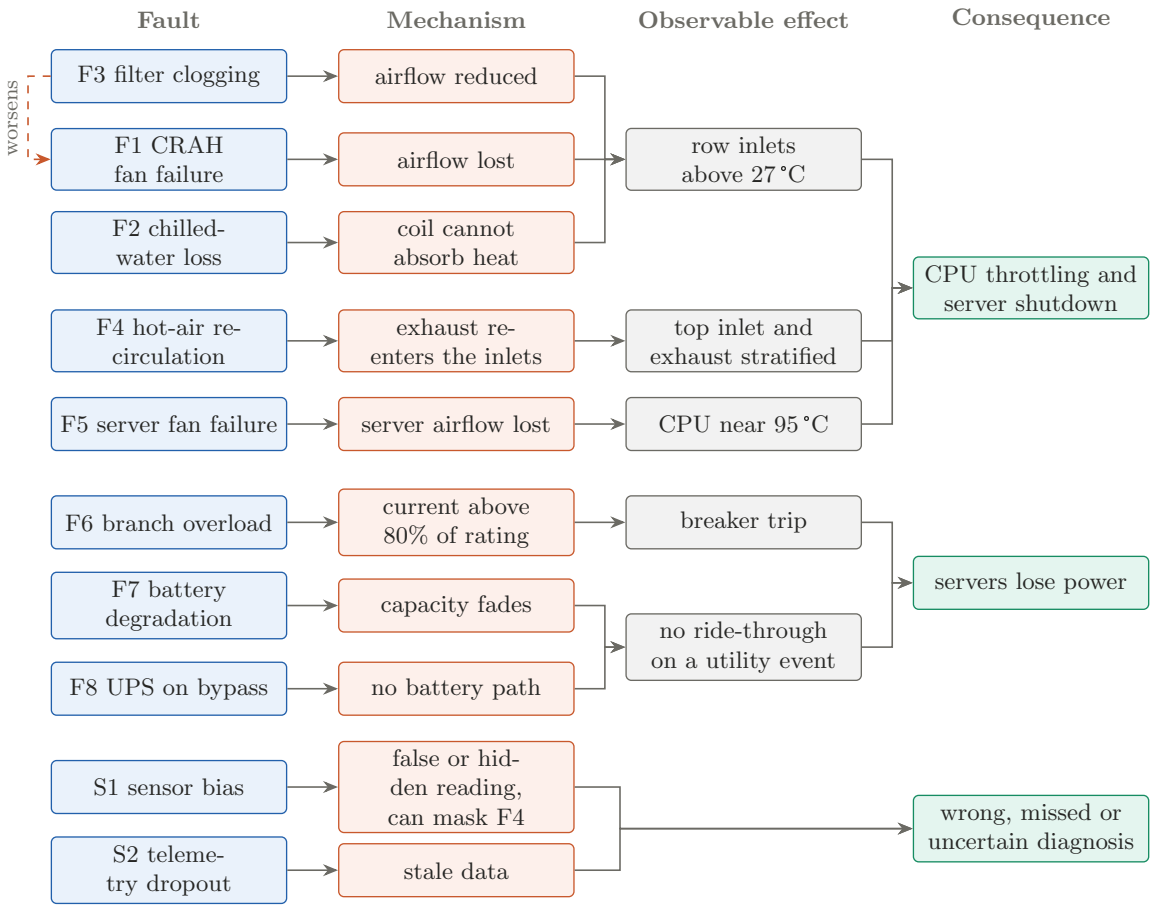
fig_architecture



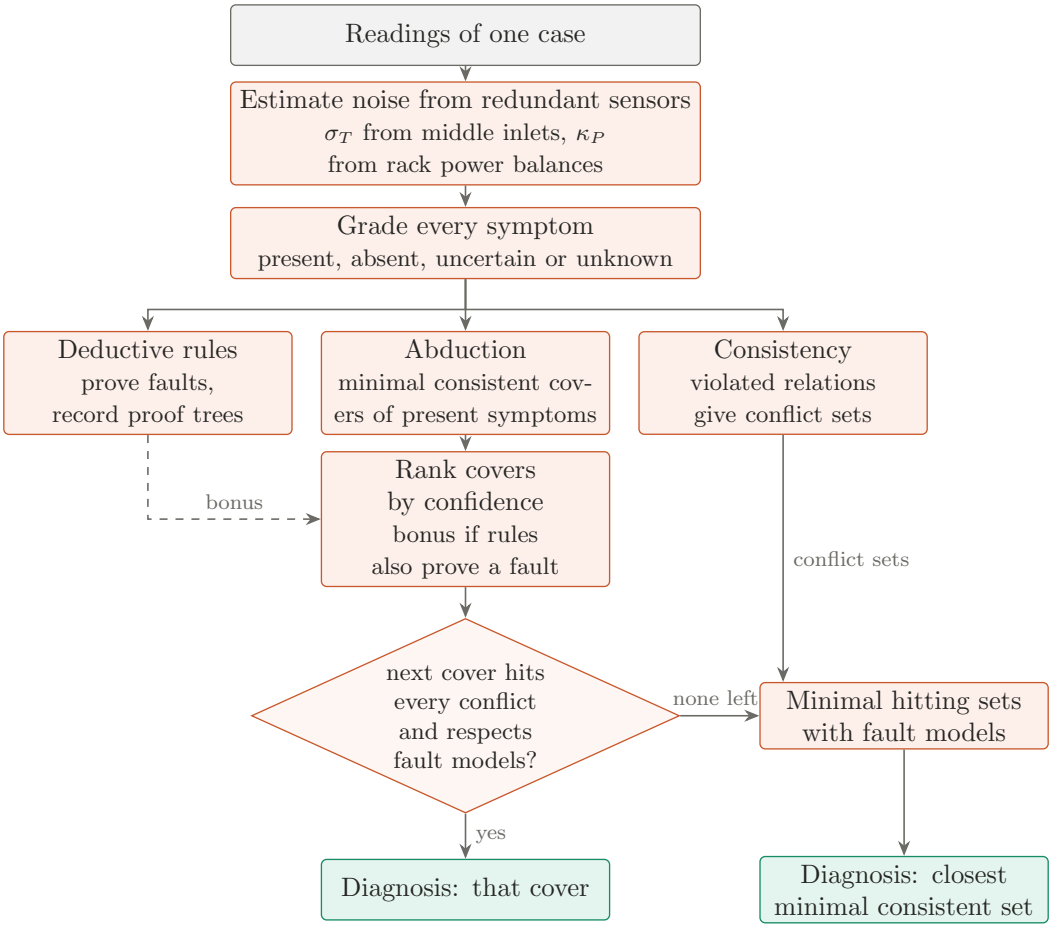
fig_knowledge_acquisition



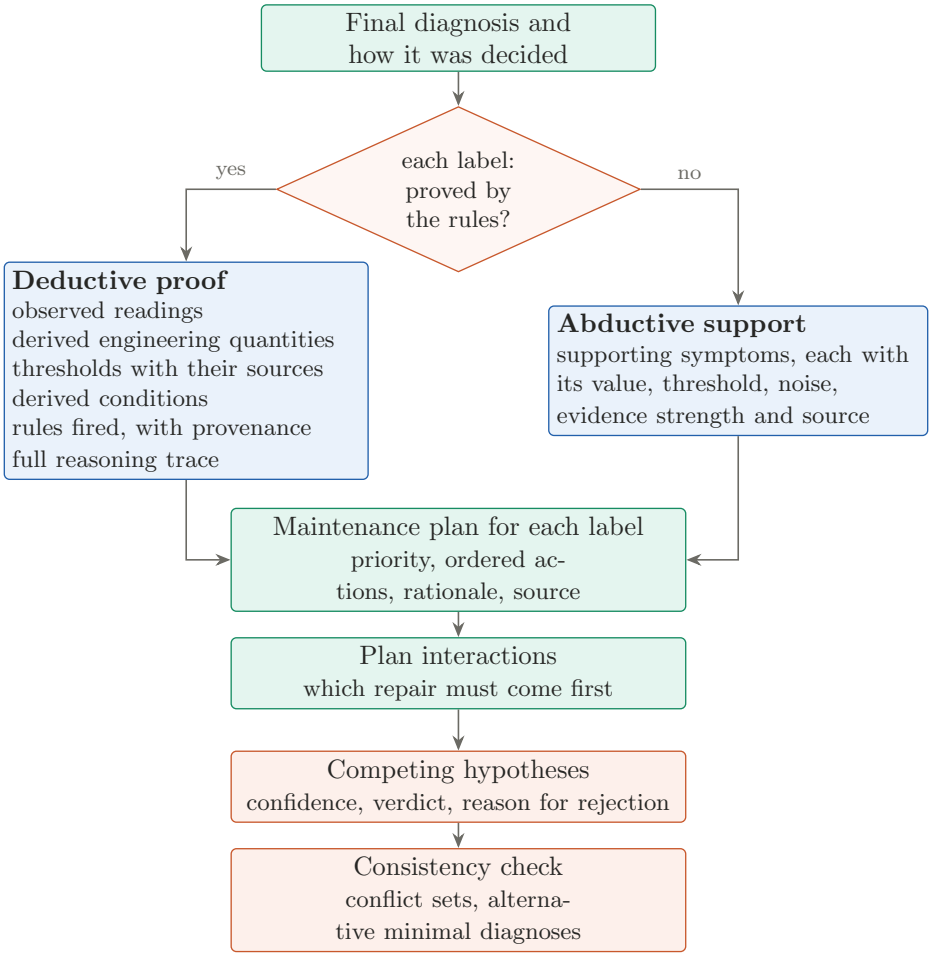
fig_fault_propagation



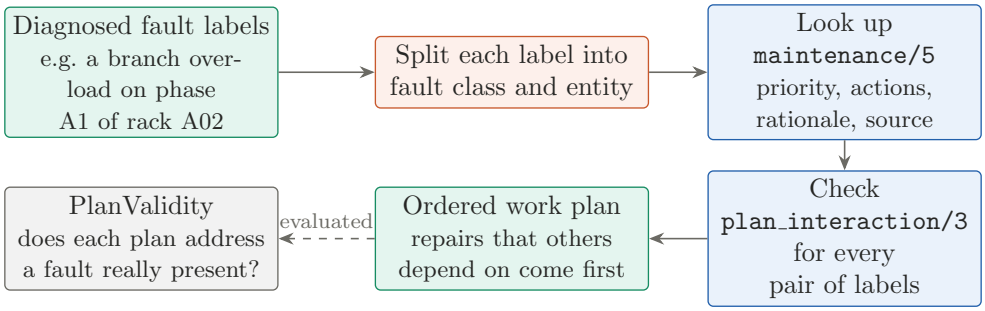
fig_diagnostic_workflow



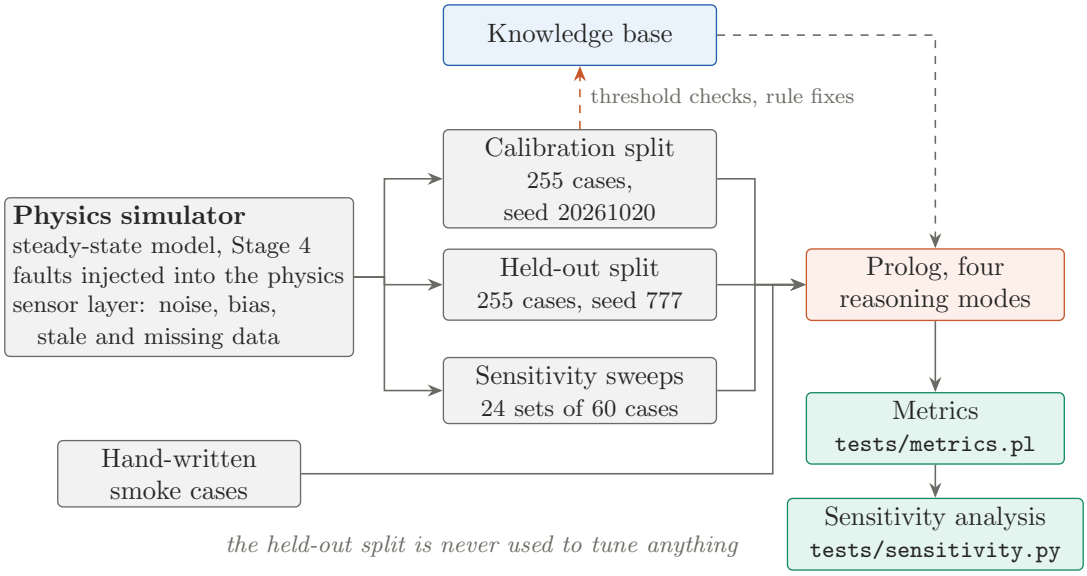
fig_explanation_workflow



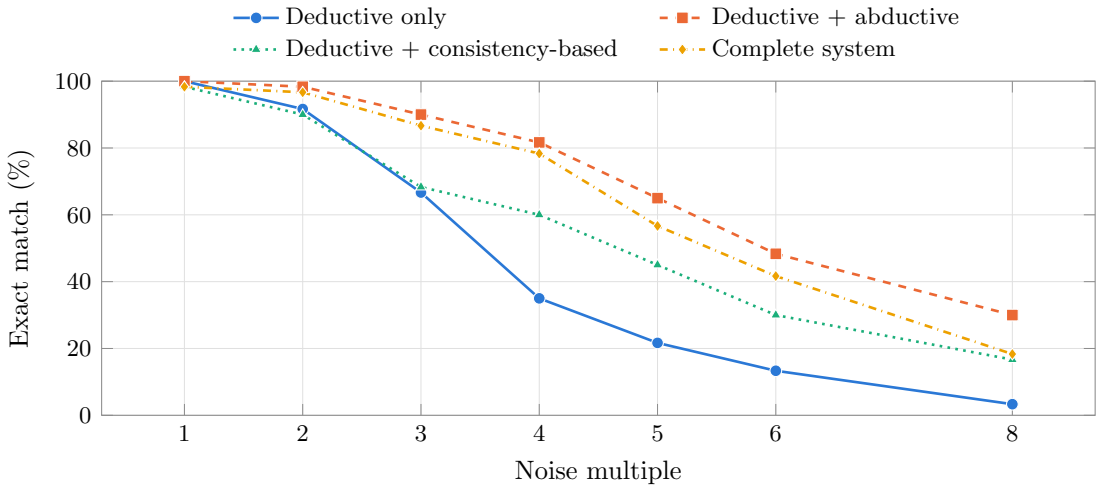
fig_maintenance_workflow



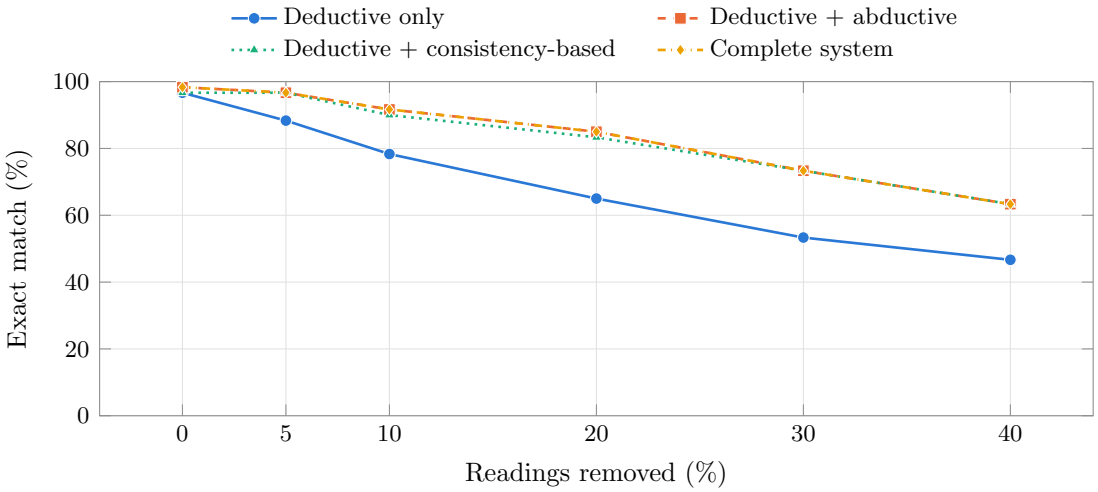
fig_evaluation_workflow



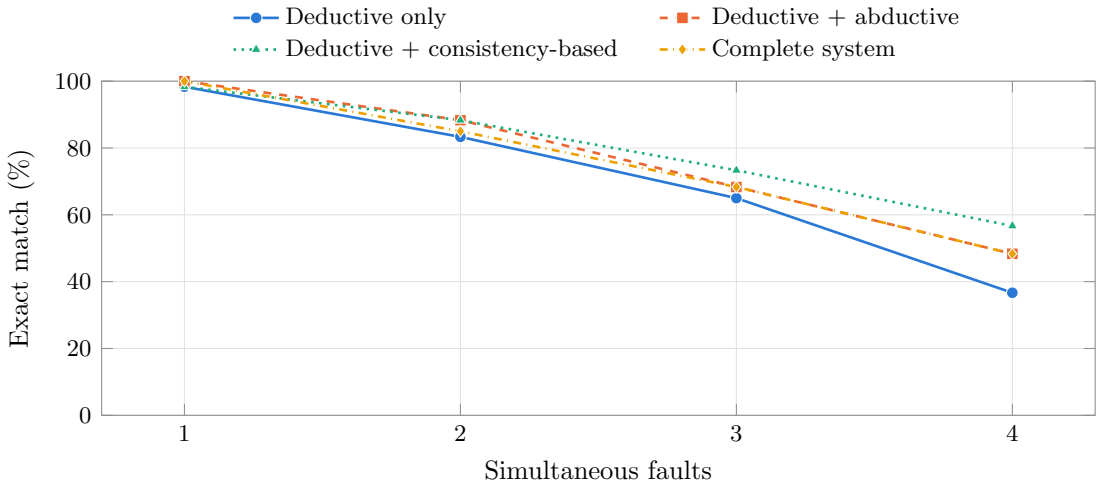
sens_noise


Deductive only
Deductive + abductive
Deductive + consistency-based
Complete system
Exact match (%)
100
80
60
40
20
0
1
2
3
4
5
6
8
Noise multiple

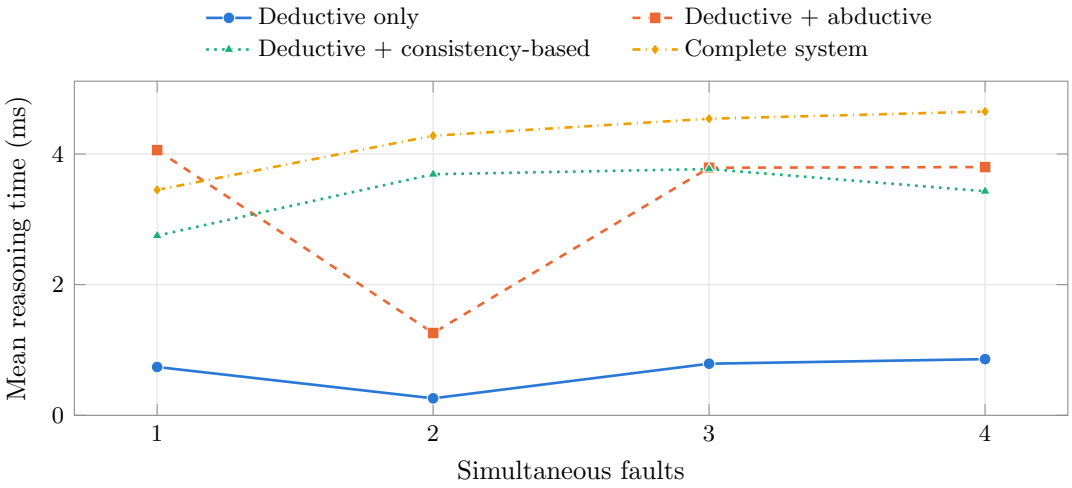
sens_missing



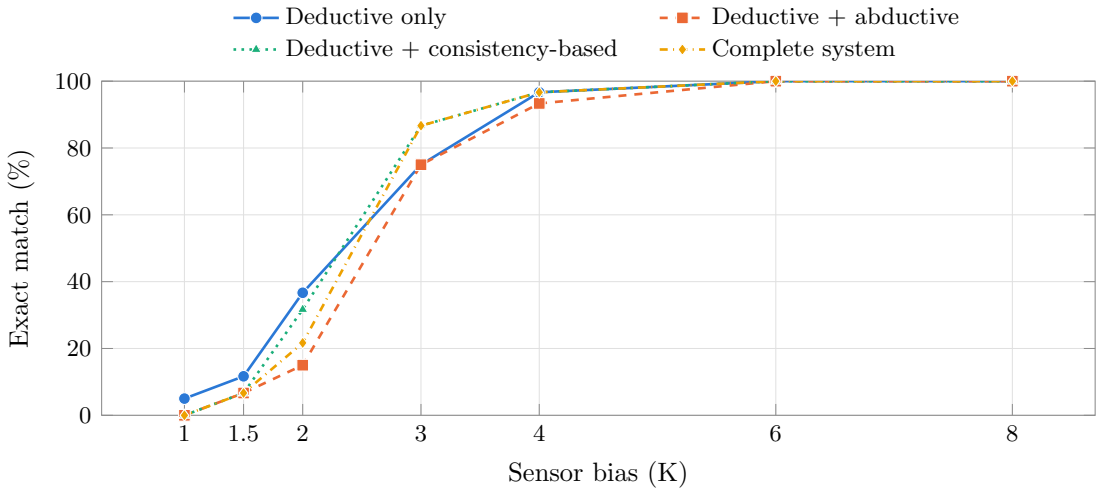
sens_multi


Deductive only
Deductive + abductive
Deductive + consistency-based
Complete system
Exact match (%)
100
80
60
40
20
0
1
2
3
4
Simultaneous faults

sens_multi_time



sens_bias


Deductive only
Deductive + abductive
Deductive + consistency-based
Complete system
Exact match (%)
100
80
60
40
20
0
1
1.5
2
3
4
6
8
Sensor bias (K)